



Department: Information Technology

**Bachelor of Information and
Communications Technology (BINCTI) & Advanced
Diploma in Information and Communications
Technology (ADICTI)**

MODULE / SUBJECT CODE: PBDV30I | PBDE40I

Project 3 – Game Application

Due Date – Refer to the study guide.

General Information and Project Task

Collaborative projects afford students the opportunity to fully engage in their learning by communicating their needs, ideas, and opinions. This not only helps to cultivate competencies in various areas of software and application development but also prepares them for future employment opportunities and other projects.

Establishment of the Project Team

This is a group project evaluation. The project teams, consisting of 10–13 members per group, were established by the lecturer based on the responses gathered from the survey administered to students. The survey results were used to ensure balanced representation of diversity and inclusiveness in terms of backgrounds, perspectives, and capabilities within each group.

Each group is required to provide the following information:

1. **Group Name:** State a unique name that best represents your group.
2. **Group Leader:** Indicate the elected group leader who will coordinate team activities and communication.

Each group member must:

- Contribute to overall project objectives.
- Complete individual deliverables.
- Provide expertise.
- Work respectfully with other group members.
- Find solutions to problems and not make excuses.

Note that a member of one group may not be allowed to migrate to another group. Any unforeseen challenges should be discussed with the lecturer. While this project emphasizes group work, it is the responsibility of each individual student to ensure that they understand the curriculum for this module and the requirements of the project.

Game Application Development Project

Objective

The objective of this project is to design and develop a fully functional 2D or 3D game using a popular game engine or framework. Students will gain practical experience in game development while demonstrating their creativity, programming skills, and understanding of game design. The game may be developed for any suitable platform, including desktop, mobile, or web-based environments.

Students may choose any game genre and are permitted to use ready-made templates, plugins, asset packs, libraries, animations, sound effects, music, APIs, and other third-party resources to accelerate development. Students are also encouraged to make use of online tutorials, official documentation, and community forums related to their chosen game engine for additional guidance and support during the development process. However, the final project must demonstrate meaningful customization and original development work.



A. Game Engine Options

Students may choose one of the following game engines or frameworks to develop their game application:

- Unity
- Unreal Engine
- Godot
- GDevelop
- Construct 3
- GameMaker
- CryEngine
- RPG Maker
- Phaser
- Pygame
- Defold
- Any other game engine or framework must be communicated and approved before using it.

B. Basic Game Requirements

The game application must meet the following requirements:

- The game must be developed using one of the approved engines or frameworks listed above.
- The game must include a clear game genre, storyline, and gameplay mechanics.
- The game may be developed as either a 2D or 3D application, depending on the project requirements.
- The game must contain at least one playable level or stage.
- The game must have a clear objective or goal for the player.
- The game must include appropriate audio and visual elements such as Sprites, Animations, Sound effects and Background music
- The game must incorporate at least two gameplay mechanics or features, for example:
 - Power-ups
 - Enemy AI
 - Physics-based interactions
 - Scoring systems
 - Health systems

Game Project Assessment Criteria

1. Concept and Originality (15 Marks)

Assessment will consider:

- Uniqueness and creativity of the game concept
- Innovation in gameplay mechanics or storytelling
- Original features or elements that make the game stand out

2. Game Design (20 Marks)

Assessment will consider:

- Clarity of the game objective or goal
- Effective and intuitive User Interface (UI) design
- Well-designed levels, environments, and characters
- Appropriate use of aesthetics, animations, and sound design

3. Visuals and Audio (15 Marks)

Assessment will consider:

- Quality and visual appeal of graphics and visual elements
- Effective use of audio to enhance the gaming experience
- Consistency and implementation of sprites, animations, and visual effects

4. Functionality and Gameplay (20 Marks)

Assessment will consider:

- Smooth and responsive controls
- Engaging and entertaining gameplay mechanics
- Clear objectives and progression systems
- Appropriate level of challenge for the player

5. Overall Execution (25 Marks)

Assessment will consider:

- Stability of the game (free from bugs and crashes)
- Overall game performance and smooth execution
- Efficient use of the selected game engine's features and capabilities
- Correct implementation of game mechanics and rules
- Effective use of scripting or programming for game logic

D. Project Documentation Requirements

Students must submit a written report that includes:

- Game overview (concept, genre, and objectives)
- Gameplay description and controls
- Tools and technologies used (game engine, plugins, assets, etc.)
- Development process (how the game was designed and built)
- Challenges encountered and how they were resolved
- References and acknowledgements (including tutorials, documentation, and third-party resources used)

E. Presentation and Submission

All group members must be present during the presentation, and at least three members are required to present. **All participants must keep their cameras on for the duration of the session.** This is essential for fair assessment and to ensure equal participation across the group. Failure to comply with these requirements will result in a score of zero for the entire project.

- **Duration** - The total duration of the presentation is set at **20 minutes**.
 - 10 minutes - For demonstrating the game features
 - 10 minutes - For discussing the code and development process.
- **Scheduling** - Each group must coordinate internally to select a suitable time for your presentation via Microsoft Teams
- **Recording** - Ensure that your presentation is recorded using the recording function in Microsoft Teams and audible enough.
- **Submission** - After completing the presentation, please submit your work by sharing a link to the video recording and the project documentation via email to freedomk@dut.ac.za no later than **4pm on 1 June 2026**.